SPATIAL REDISTRIBUTION OF PREDATOR-PREY DYNAMICS

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ABSTRACT

This poster presents two predator–prey models in a two-patch environment. In this study, we will work with a system of four ordinary differential equations that govern the densities of the prey and predator populations on each patch. Here, we assume that migration between patches is faster than predator–prey interactions and other phenomena. Generally, prey leave a patch at a migration rate proportional to the density of predators in that patch. Predators leave a patch at a migration rate inversely proportional to the density of the local prey population in that patch. We take advantage of two time scales to reduce the complete model to a system of two equations governing the total prey and predator densities. In the study, the stability analysis of the aggregated model shows that a unique strictly positive equilibrium exists. This equilibrium may be stable or unstable. A Hopf bifurcation may occur, leading the equilibrium to be a centre. The theoretical analysis is followed by numerical simulations using MATLAB to visualize the models by phase portraits and time-series plots.

COMPLETE PREY-PREDATOR MODEL

The complete system, over the rapid time scale $\tau = t/\varepsilon$, is as follows:

$dn_1/d\tau = (m'(p_2)\cdot n_2 - m(p_1)\cdot n_1) + \varepsilon[r_1\cdot n_1 - a_1\cdot n_1\cdot p_1]$,
$dn_2/d\tau = (m(p_1)\cdot n_1 - m'(p_2)\cdot n_2) + \varepsilon[r_2\cdot n_2 - a_2\cdot n_2\cdot p_2]$,
$dp_1/d\tau = (k'\cdot p_2 - k\cdot p_1) + \varepsilon[-m_1\cdot p_1 + b_1\cdot n_1\cdot p_1]$,
$dp_2/d\tau = (k\cdot p_1 - k'\cdot p_2) + \varepsilon[-m_2\cdot p_2 + b_2\cdot n_2\cdot p_2]$,

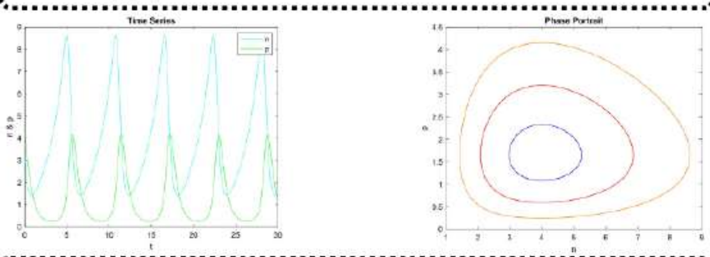
where,
$n_1(t)$ & $n_2(t)$ = prey population densities at time t, in patch 1 and 2, respectively.
$p_1(t)$ & $p_2(t)$ = predator population densities at time t, in patch 1 and 2, respectively.
r_i = intrinsic growth rate of the prey population in patch i (i=1,2), where $r_1 \neq r_2$.
a_i & b_i = predation parameters on patch i (i=1,2).
m_i = natural mortality rate of the predator in patch i.
k & k' = predator migration rates from patch 1 to patch 2 and vice versa.
The prey migration rates are predator density dependent as follows:
$m(p_1) = \alpha\cdot p_1 + \alpha_0$
$m'(p_2) = \beta\cdot p_2 + \alpha_0$
Here, α_0, α and β are positive parameters, and ε is a small dimensionless parameter.

AGGREGATED PREY-PREDATOR MODEL

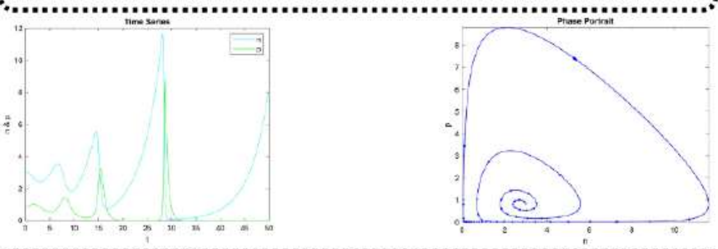
The aggregated system is given by:
$dn/dt = (1 / (\delta p + 2\alpha_0)) * (rn + anp - \tilde{b}np^2)$,
$dp/dt = -Mp + (1 / (\delta p + 2\alpha_0)) * (bnp + cnp^2)$,

Where the parameters are defined as:
$r = \alpha_0(r_1 + r_2)$,
$a = r_1\beta\mu_2 + r_2\alpha\mu_1 - \alpha_0(a_1\mu_1 + a_2\mu_2)$,
$\tilde{b} = \mu_1\mu_2(a_1\beta + a_2\alpha)$,
$M = m_1\mu_1 + m_2\mu_2$,
$b = \alpha_0(b_1\mu_1 + b_2\mu_2)$,
$c = \mu_1\mu_2(b_1\beta + b_2\alpha)$.

Parameters have been chosen as: $\delta = 2, \alpha_0 = 1$, r = 0.5, a = 3, $\tilde{b}$ = 2, M = 2, b = 1 and c = 0.5. (Q<0)



Parameters have been chosen as: $\delta = 2, \alpha_0 = 1$, r = 0.5, a = 3, $\tilde{b}$ = 2, M = 2, b = 1 and c = 1. (Q=0)



Parameters have been chosen as: $\delta = 2, \alpha_0 = 1$, r = 0.5, a = 1, $\tilde{b}$ = 2, M = 2, b = 1 and c = 2. (Q>0)

STABILITY AND STABILITY PARAMETER

In this study, we have encountered one parameter $B := (p^* \times M \times (2\alpha_0 c - \delta b)) / ((b + cp)(\delta p^* + 2\alpha_0))$, whose change in value alters the stability of the system. Moreover, the stability depends on $Q = 2\alpha_0 c - \delta b$. The details are given in the following table:

Value of Q	Stability
Q < 0	Stable focus
Q = 0	Centre
Q > 0	Unstable focus

SUGGESTIONS

· The case, where, $m(p_1) = \alpha\cdot p_1^h + \alpha_0$ and $m(p_2) = \beta\cdot p_2^h + \alpha_0$, with h > 1, needs to be studied further.

· An interesting future direction would be the study of a repulsive–attractive model, where, when predator density is high on a patch, prey disperse more (repulsion); when prey density is high, predators remain on that patch; otherwise, they leave (attraction).